

WEB DESIGN AND DEVELOPMENT COURSE OUTLINE

2-Month Certification Program

Institution: Masomotele Technical Training Institute (M.T.T.I)

Duration: 8 Weeks (2 Months)

Motto: Start Learning, Start Earning

COURSE OVERVIEW

This comprehensive 2-month program is designed to equip students with essential web design and development skills. Students will learn to create responsive, modern websites using HTML, CSS, and JavaScript, as well as build professional websites using WordPress. Upon completion, graduates will be able to design and develop functional websites for personal projects, businesses, or clients.

COURSE OBJECTIVES

By the end of this course, students will be able to:

- Understand the fundamentals of web design principles
 - Create well-structured web pages using HTML5
 - Apply basic styling and layouts using CSS3
 - Add simple interactivity to websites using basic JavaScript
 - Build and customize professional websites using WordPress
 - Deploy and maintain websites online
 - Create a professional portfolio of web projects
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WEEK 1: INTRODUCTION TO WEB DESIGN & HTML BASICS

Day 1-2: Introduction to Web Development

- What is Web Design vs Web Development?
- How the Internet and Web work

- Understanding domains, hosting, and servers
- Overview of web technologies (HTML, CSS, JavaScript)
- Setting up development environment (VS Code, browsers, extensions)
- Introduction to developer tools

Day 3-4: HTML Fundamentals

- HTML structure and syntax
- Basic HTML tags (headings, paragraphs, links, images)
- HTML5 semantic elements
- Creating your first web page
- Understanding document structure (DOCTYPE, head, body)

Day 5-6: HTML Content Elements

- Lists (ordered, unordered, definition)
- Tables and table formatting
- Forms and input elements
- Form validation attributes
- Embedding media (audio, video, iframe)

Week 1 Project: Create a multi-page personal portfolio website using HTML

WEEK 2: ADVANCED HTML & INTRODUCTION TO CSS

Day 1-2: Advanced HTML Concepts

- HTML5 semantic tags (header, footer, nav, article, section)
- Accessibility best practices
- Meta tags and SEO basics
- Special characters and entities
- HTML best practices and code organization

Day 3-4: CSS Basics

- What is CSS and how it works
- CSS syntax and selectors
- Inline, internal, and external CSS
- Colors, backgrounds, and gradients
- Text styling and web fonts (Google Fonts)
- The CSS box model

Day 5-6: CSS Layout Fundamentals

- Display properties
- Positioning (static, relative, absolute, fixed, sticky)
- Float and clear
- Understanding margins, padding, and borders
- Width and height properties

Week 2 Project: Style the personal portfolio with CSS

WEEK 3: BASIC CSS PRACTICE & INTRODUCTION TO RESPONSIVE DESIGN

Day 1-2: CSS Practice & Simple Layouts

- Creating navigation menus with CSS
- Styling forms and buttons
- Working with borders and shadows
- Creating simple page layouts
- Float-based layouts
- Building multi-column layouts

Day 3-4: Introduction to Responsive Design

- What is responsive design?
- Using percentages for flexible layouts
- Basic media queries
- Making images responsive
- Viewport meta tag
- Testing on different screen sizes

Day 5-6: CSS Frameworks Introduction

- What are CSS frameworks?
- Introduction to Bootstrap basics
- Using Bootstrap grid system
- Pre-styled Bootstrap components
- Customizing Bootstrap colors
- Creating a simple Bootstrap page

Week 3 Project: Build a simple multi-page website with basic responsive features

WEEK 4: INTRODUCTION TO JAVASCRIPT BASICS

Day 1-2: JavaScript Fundamentals

- What is JavaScript and what can it do?
- Adding JavaScript to web pages (inline, internal, external)
- Variables and data types (strings, numbers, booleans)
- Basic operators (arithmetic, assignment)
- Console.log for testing
- Comments in JavaScript
- Simple input/output (alert, prompt)

Day 3-4: Basic JavaScript Operations

- Simple calculations
- Working with strings (concatenation, length)
- Basic conditional statements (if, else)
- Comparison operators (==, !=, >, <)
- Simple examples and exercises
- Debugging basics

Day 5-6: Basic Interactivity

- Introduction to functions (simple examples)
- Onclick events
- Changing text content with JavaScript
- Show/hide elements
- Simple image changes
- Basic form interactions

Week 4 Project: Create a simple interactive page (e.g., button clicks change content, simple calculator)

WEEK 5: BASIC JAVASCRIPT DOM & WEB DESIGN PRINCIPLES

Day 1-2: Simple DOM Manipulation

- What is the DOM?
- Selecting elements by ID
- Changing text and HTML content
- Modifying styles with JavaScript

- Adding and removing classes
- Simple practical examples

Day 3-4: Basic Events & Form Validation

- Common events (click, submit)
- Basic event listeners
- Simple form validation (checking empty fields)
- Displaying messages to users
- Enabling/disabling buttons
- Practical form examples

Day 5-6: Web Design Principles & Portfolio Preparation

- Color theory basics
- Typography and readability
- Layout principles and white space
- User experience (UX) basics
- Creating effective navigation
- Preparing content for WordPress
- Planning website structure

Week 5 Project: Create a complete HTML/CSS portfolio site with basic JavaScript interactions (form validation, menu toggle)

WEEK 6: INTRODUCTION TO WORDPRESS

Day 1-2: WordPress Basics

- What is WordPress and why use it?
- WordPress.com vs WordPress.org
- Setting up local development environment (XAMPP/Local)
- Installing WordPress
- WordPress dashboard overview
- Basic settings and configurations

Day 3-4: WordPress Content Management

- Posts vs Pages
- Categories and tags
- Media library management
- Creating and editing content
- The WordPress editor (Gutenberg)

- Permalinks and SEO-friendly URLs
- Managing comments

Day 5-6: WordPress Themes

- Understanding WordPress themes
- Installing and activating themes
- Free vs premium themes
- Customizing themes with Customizer
- Working with widgets and menus
- Child themes concept
- Popular free themes (Astra, OceanWP, GeneratePress)

Week 6 Project: Create a blog website using WordPress

WEEK 7: WORDPRESS CUSTOMIZATION & PLUGINS

Day 1-2: WordPress Plugins

- What are plugins and how they work
- Installing and managing plugins
- Essential plugins overview:
 - Contact forms (Contact Form 7, WPForms)
 - SEO (Yoast SEO)
 - Security (Wordfence)
 - Backup (UpdraftPlus)
 - Performance (WP Super Cache)
- Plugin best practices

Day 3-4: Page Builders

- Introduction to page builders
- Installing Elementor (free version)
- Elementor interface and widgets
- Building pages with drag and drop
- Creating custom layouts
- Responsive design in Elementor
- Pre-built templates

Day 5-6: E-commerce Basics

- Introduction to WooCommerce
- Setting up an online store

- Adding products and categories
- Payment gateway setup (M-Pesa integration)
- Shipping options
- Basic store customization

Week 7 Project: Build a small business website with contact form and services pages

WEEK 8: FINAL PROJECT & DEPLOYMENT

Day 1-2: Website Deployment

- Understanding web hosting
- Choosing hosting providers (local options: Safaricom, Truhost, etc.)
- Domain registration in Kenya
- Uploading files via FTP/cPanel
- Deploying WordPress sites
- Moving WordPress from local to live
- Database management basics

Day 3-4: Website Optimization & Maintenance

- Website speed optimization
- Image optimization
- Basic SEO implementation
- Google Analytics setup
- Website security best practices
- Regular backups
- Update management
- Troubleshooting common issues

Day 5-6: Final Project Completion

- Students work on comprehensive final project
- One-on-one consultations
- Project presentations
- Portfolio review
- Course wrap-up and next steps
- Freelancing basics and finding clients in Kenya
- Continuing education resources

Week 8 Final Project: Students choose one:

1. Complete business website with WordPress (5+ pages, contact form, blog)

2. E-commerce store with at least 10 products
 3. Custom HTML/CSS/JS website with interactive features
 4. Portfolio website showcasing all course projects
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ASSESSMENT METHODS

1. **Weekly Practical Assignments (40%)**
 - Hands-on coding exercises
 - Mini-projects at end of each week
 2. **Mid-Course Project (20%)**
 - Week 5: Complete portfolio website with HTML, CSS, and basic JavaScript
 3. **Final Project (30%)**
 - Week 8: Comprehensive website project
 4. **Class Participation & Attendance (10%)**
 - Active participation in class
 - Completion of daily exercises
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REQUIRED TOOLS & SOFTWARE

Free Software Required:

- Text Editor: Visual Studio Code
- Web Browsers: Chrome, Firefox
- Local Server: XAMPP or Local by Flywheel
- FTP Client: FileZilla
- Image Editor: GIMP or Canva (free version)
- Version Control: Git (optional, for advanced students)

Recommended Resources:

- HTML, CSS, JavaScript reference: MDN Web Docs
 - WordPress documentation: WordPress.org
 - Practice platforms: CodePen, JSFiddle
 - Free images: Unsplash, Pexels
 - Icons: Font Awesome, Flaticon
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COURSE MATERIALS PROVIDED

- Comprehensive printed notes for each week

- Code examples and templates
 - Project starter files
 - Checklists and quick reference guides
 - Access to recorded sessions (if available)
 - Sample websites for practice
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PREREQUISITES

- Basic computer literacy
 - Ability to browse the internet
 - No prior programming experience required
 - Enthusiasm to learn and create!
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CAREER OPPORTUNITIES AFTER COMPLETION

Graduates of this course can pursue:

- Freelance Web Designer/Developer
 - Junior Front-End Developer
 - WordPress Developer
 - Web Content Manager
 - UI/UX Designer (with additional training)
 - Digital Marketing Specialist (web focus)
 - Start a web design agency
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CERTIFICATION

Upon successful completion of the course with a passing grade of 70% or above, students will receive:

- **Certificate of Completion in Web Design and Development**
 - Issued by Masomotele Technical Training Institute (M.T.T.I)
 - TVETA-accredited certification
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COURSE SCHEDULE

- **Duration:** 8 Weeks (2 Months)
 - **Class Days:** Monday to Friday
 - **Class Time:** [To be confirmed - typically 2-3 hours per day]
 - **Total Hours:** 80-100 contact hours
 - **Format:** Hands-on practical training with theory
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STUDENT SUCCESS TIPS

1. **Practice Daily:** Code every day, even if just for 30 minutes
 2. **Build Projects:** Apply what you learn immediately
 3. **Use Resources:** Don't hesitate to Google and use documentation
 4. **Ask Questions:** No question is too basic
 5. **Collaborate:** Work with classmates on projects
 6. **Build Portfolio:** Document all your projects
 7. **Stay Updated:** Follow web development blogs and channels
 8. **Network:** Connect with local developers and attend meetups
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INSTRUCTOR SUPPORT

- In-class demonstrations and guidance
 - One-on-one help during lab sessions
 - [Email/WhatsApp support] for urgent questions
 - Weekly office hours for additional help
 - Career guidance and mentorship
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NEXT STEPS AFTER COMPLETION

Students can advance their skills with:

- **Advanced CSS** (Flexbox, Grid, Animations, Sass)
 - **Advanced JavaScript** (ES6+, APIs, Frameworks like React)
 - **Backend Development** (PHP and MySQL)
 - **Advanced WordPress Development** (Custom themes and plugins)
 - **Graphic Design and UI/UX**
 - **Digital Marketing specialization**
 - **E-commerce Specialization** (WooCommerce mastery)
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CONTACT INFORMATION

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This course outline is subject to minor adjustments based on class progress and student needs. M.T.T.I reserves the right to update the curriculum to reflect current industry standards and best practices.

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